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PODS

EXPERIMENT 2 - DATA DISCOVERY & PREPARATION

Learning Objectives

The objective of this experiment is to understand the process of data discovery and preparation by exploring a customer churn dataset. It aims to learn how to clean and preprocess data, perform distribution and relationship analysis, detect outliers, prepare the dataset using train-test splitting, and save the processed data for machine learning applications.

Libraries Used

This experiment uses the **Pandas** library for data manipulation and preprocessing, **Matplotlib** and **Seaborn** for creating statistical visualizations, **LabelEncoder** to convert categorical variables into numerical values, **StandardScaler** to standardize numerical features, and **train_test_split** from Scikit-learn to divide the dataset into training and testing sets.

coding

```
import pandas as pd import

matplotlib.pyplot as plt

import seaborn as sns


from sklearn.preprocessing import LabelEncoder, StandardScaler

from sklearn.model_selection import train_test_split

df =
pd.read_csv("C:\\Users\\thirunesh\\Downloads\\archive\\customer_churn_datasettesting-
master.csv")
```

```
print("First 10 Records")
print(df.head(10))

print("\nLast 10 Records")
print(df.tail(10))

print("\nShape")
print(df.shape)

print("\nColumns")
print(df.columns)


print("\nDataset Info")
print(df.info())

print("\nSummary")
print(df.describe())

print("\nMissing Values")
print(df.isnull().sum())

print("\nDuplicate Records")
print(df.duplicated().sum())
# Step 4 : Data Cleaning


df.drop_duplicates(inplace=True)
```

```
for col in df.columns:

    if df[col].dtype == "object":
        df[col] = df[col].fillna(df[col].mode()[0])

    else:
        df[col] = df[col].fillna(df[col].median())

print("\nMissing Values After Cleaning")
print(df.isnull().sum())



encoder = LabelEncoder()

categorical = [
    "Gender",
    "Subscription Type",
    "Contract Length",
    "Churn"
]

for col in categorical:
    if col in df.columns:
        df[col] = encoder.fit_transform(df[col])
```

```
scaler = StandardScaler()


num_cols = [
    "Age",
    "Tenure",
    "Usage Frequency",
    "Support Calls",
    "Payment Delay", "Total
    Spend",
    "Last Interaction"
]


df[num_cols] = scaler.fit_transform(df[num_cols])




plt.figure(figsize=(10,8))


sns.heatmap(df.corr(numeric_only=True),
            annot=True,
            cmap="coolwarm")


plt.title("Correlation Heatmap")
plt.show()




cols = [
```

```
    "Churn",
    "Gender",
    "Contract Length",
    "Subscription Type"
]

for col in cols:

    plt.figure(figsize=(6,4))

    sns.countplot(x=df[col])

    plt.title(col)

    plt.show()




plot_cols = [
    "Age",
    "Tenure",
    "Total Spend"
]
for col in plot_cols:

    plt.figure(figsize=(6,4))

    sns.histplot(df[col], kde=True)
```

```
    plt.title("Histogram of " + col)

    plt.show()

    plt.figure(figsize=(6,4))

    sns.boxplot(x=df[col])

    plt.title("Boxplot of " + col)

    plt.show()

print("\nOutliers using IQR")

for col in num_cols:

    Q1 = df[col].quantile(0.25)

    Q3 = df[col].quantile(0.75)

    IQR = Q3 - Q1

    lower = Q1 - 1.5 * IQR
    upper = Q3 + 1.5 * IQR

    outliers = df[(df[col] < lower) | (df[col] > upper)]
```

```
    print(col, ":", len(outliers))


plt.figure(figsize=(6,4))
sns.countplot(x="Contract Length", hue="Churn", data=df)
plt.title("Contract Length vs Churn")
plt.show()


plt.figure(figsize=(6,4))
sns.countplot(x="Subscription Type", hue="Churn", data=df)
plt.title("Subscription Type vs Churn")
plt.show()


plt.figure(figsize=(6,4))
sns.boxplot(x="Churn", y="Total Spend", data=df) plt.title("Total
Spend vs Churn")
plt.show()


plt.figure(figsize=(6,4))
sns.boxplot(x="Churn", y="Tenure", data=df) plt.title("Tenure
vs Churn")
plt.show()



plt.figure(figsize=(6,4))
sns.scatterplot(x="Tenure",
                y="Total Spend",
                hue="Churn",
                data=df)
```

```
plt.title("Tenure vs Total Spend")

plt.show()

print("\nContract Length with Highest Churn")

print(df.groupby("Contract Length")["Churn"].mean())

print("\nSubscription Type with Highest Churn")

print(df.groupby("Subscription Type")["Churn"].mean())

print("\nAverage Tenure by Churn")

print(df.groupby("Churn")["Tenure"].mean())

print("\nAverage Total Spend by Churn")

print(df.groupby("Churn")["Total Spend"].mean())

print("\nCorrelation with Churn")

print(df.corr(numeric_only=True)["Churn"].sort_values(ascending=False))          X          =
df.drop("Churn", axis=1)

y = df["Churn"]
```

```
X_train, X_test, y_train, y_test =
    train_test_split( X, y, test_size=0.20,
    random_state=42
)

print("\nTraining Data Shape :", X_train.shape)

print("Testing Data Shape :", X_test.shape)

df.to_csv("CustomerChurn_Preprocessed.csv", index=False)

print("\nDataset Saved Successfully!")

print("\nFile Name : CustomerChurn_Preprocessed.csv")
```

OUTPUT:

First 10 Records

	CustomerID	Age	Gender	Tenure	Usage Frequency	Support Calls \
0	1	22	Female	25	14	4
1	2	41	Female	28	28	7
2	3	47	Male	27	10	2
3	4	35	Male	9	12	5
4	5	53	Female	58	24	9
5	6	30	Male	41	14	10
6	7	47	Female	37	15	9
7	8	54	Female	36	11	0
8	9	36	Male	20	5	10

9 10 65 Male 8 4 2

Payment Delay Subscription Type Contract Length Total Spend \

0	27	Basic	Monthly	598
1	13	Standard	Monthly	584
2	29	Premium	Annual	757
3	17	Premium	Quarterly	232
4	2	Standard	Annual	533
5	10	Premium	Monthly	500
6	28	Basic	Quarterly	574
7	18	Standard	Monthly	323
8	8	Basic	Monthly	687
9	23	Basic	Annual	995

Last Interaction Churn

0	9	1
1	20	0
2	21	0
3	18	0
4	18	0
5	29	0
6	14	1
7	16	0
8	8	0
9	10	0

Last 10 Records

CustomerID Age Gender Tenure Usage Frequency Support Calls \

64364	64365	48	Female	36	22	4
64365	64366	26	Female	54	24	2
64366	64367	64	Male	30	27	9
64367	64368	51	Male	47	17	10
64368	64369	45	Male	3	1	5
64369	64370	45	Female	33	12	6
64370	64371	37	Male	6	1	5
64371	64372	25	Male	39	14	8
64372	64373	50	Female	18	19	7
64373	64374	52	Female	45	15	9

	Payment Delay	Subscription Type	Contract Length	Total Spend \
64364	22	Premium	Annual	342
64365	21	Basic	Annual	165
64366	28	Premium	Annual	884
64367	26	Basic	Annual	544
64368	11	Standard	Annual	218
64369	21	Basic	Quarterly	947
64370	22	Standard	Annual	923
64371	30	Premium	Monthly	327
64372	22	Standard	Monthly	540
64373	25	Standard	Monthly	696

	Last Interaction	Churn
64364	9	1
64365	23	1
64366	21	1
64367	24	1

64368 15 1

64369 14 1

64370 9 1

64371 20 1

64372 13 1

64373 22 1

Shape

(64374, 12)

Columns

Index(['CustomerID', 'Age', 'Gender', 'Tenure', 'Usage Frequency',
'Support Calls', 'Payment Delay', 'Subscription Type',
'Contract Length', 'Total Spend', 'Last Interaction', 'Churn'],
dtype='object')

Dataset Info

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 64374 entries, [1] to 64373

Data columns (total 12 columns):

Column Non-Null Count Dtype

1 Age 64374 non-null int64

2 Gender 64374 non-null object

3 Tenure 64374 non-null int64

4 Usage Frequency 64374 non-null int64

5 Support Calls 64374 non-null int64

---

[1] CustomerID 64374 non-null int64

6 Payment Delay 64374 non-null int64

7 Subscription Type 64374 non-null object

8 Contract Length 64374 non-null object

9 Total Spend 64374 non-null int64

10 Last Interaction 64374 non-null int64

11 Churn 64374 non-null int64

dtypes: int64(9), object(3) memory

usage: 5.9+ MB

None

Summary

CustomerID Age Tenure Usage Frequency \ count

64374.000000 64374.000000 64374.000000 64374.000000

mean 32187.500000 41.970982 31.994827
15.080234 std 18583.317451 13.924911
17.098234 8.816470
min 1.000000 18.000000 1.000000 1.000000

25% 16094.250000 30.000000 18.000000 7.000000

50% 32187.500000 42.000000 33.000000 15.000000

75% 48280.750000 54.000000 47.000000 23.000000
max 64374.000000 65.000000 60.000000 30.000000

Support Calls Payment Delay Total Spend Last Interaction \ count

64374.000000 64374.000000 64374.000000 64374.000000 mean

5.400690 17.133952 541.023379 15.498850 std 3.114005 8.852211

260.874809 8.638436

min 0.000000 0.000000 100.000000 1.000000

25% 3.000000 10.000000 313.000000 8.000000
50% 6.000000 19.000000 534.000000 15.000000

75%	8.000000	25.000000	768.000000	23.000000
max	10.000000	30.000000	1000.000000	30.000000

Churn

count	64374.000000
mean	0.473685
std	0.499311
min	0.000000
25%	0.000000
50%	0.000000
75%	1.000000
max	1.000000

Missing Values

CustomerID	0
Age	0
Gender	0
Tenure	0
Usage Frequency	0
Support Calls	0
Payment Delay	0
Subscription Type	0
Contract Length	0
Total Spend	0
Last Interaction	0
Churn	0 dtype: int64

Duplicate Records

0

Missing Values After Cleaning

CustomerID 0

Age 0

Gender 0

Tenure 0

Usage Frequency 0

Support Calls 0

Payment Delay 0

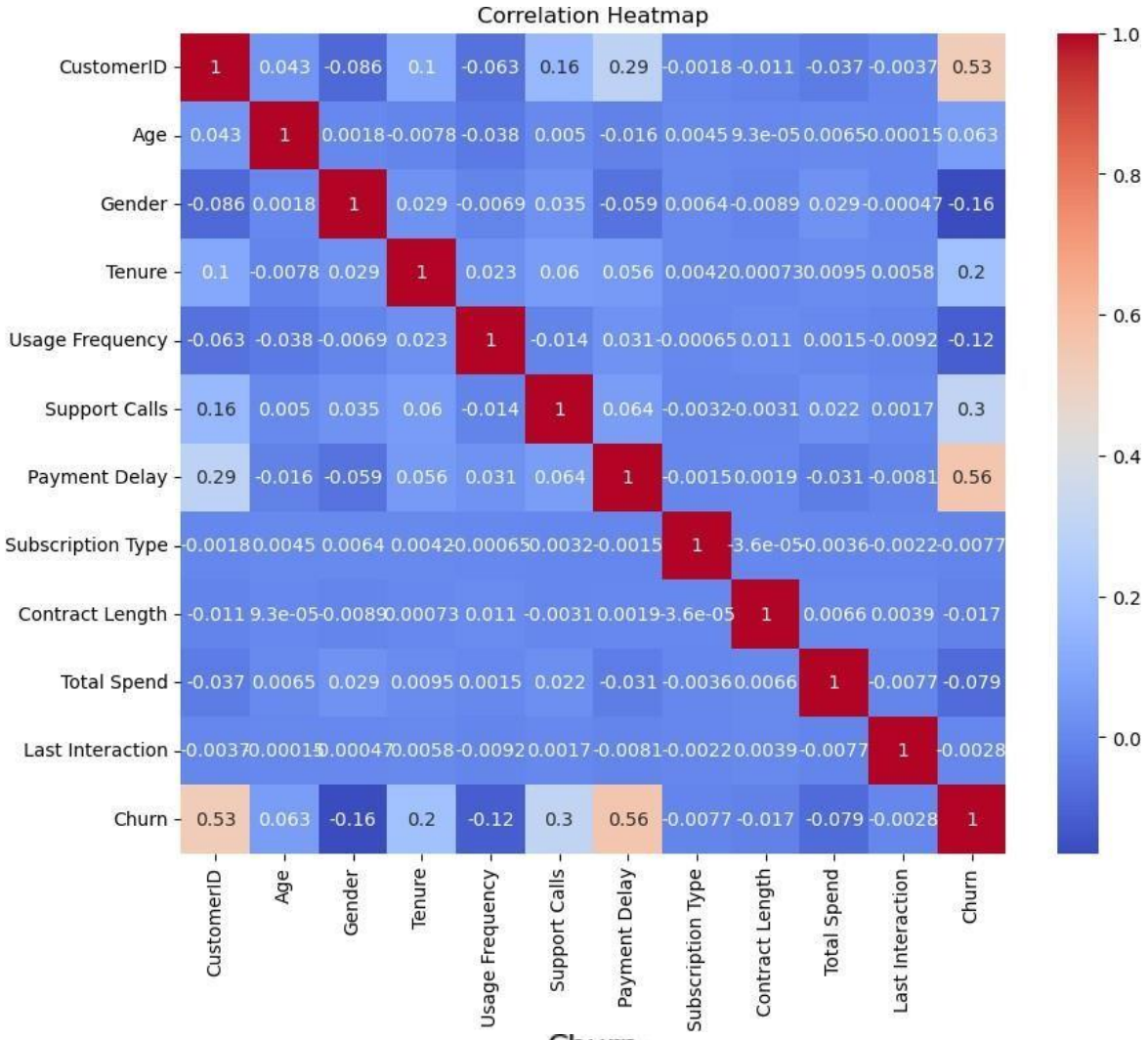
Subscription Type 0

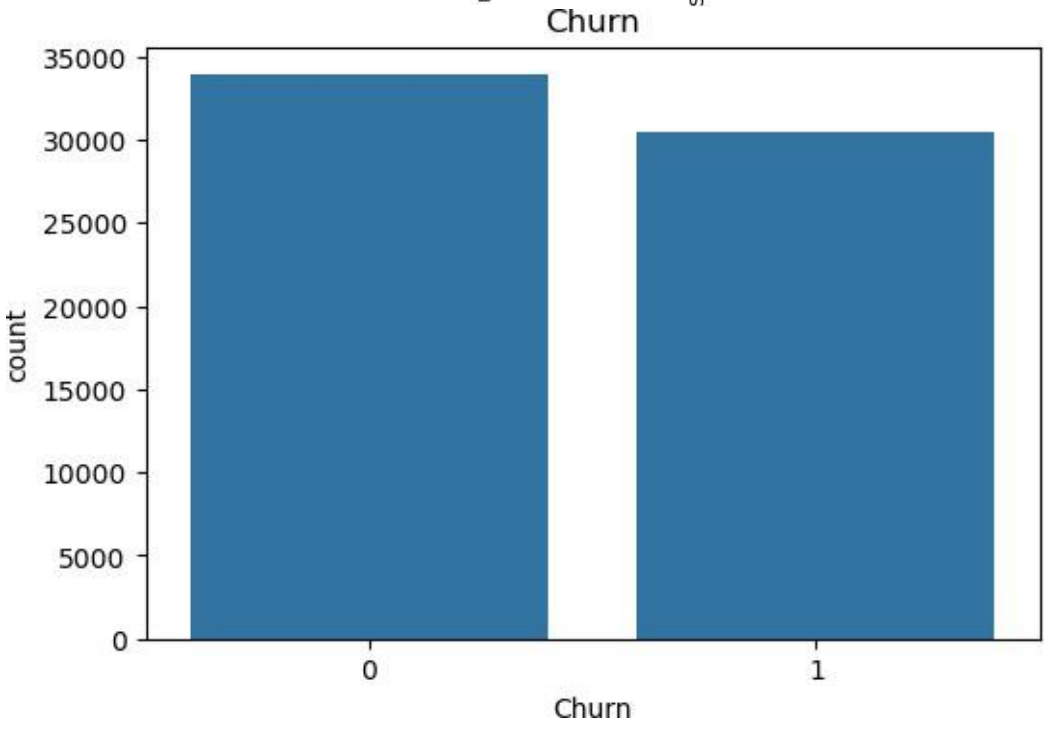
Contract Length 0

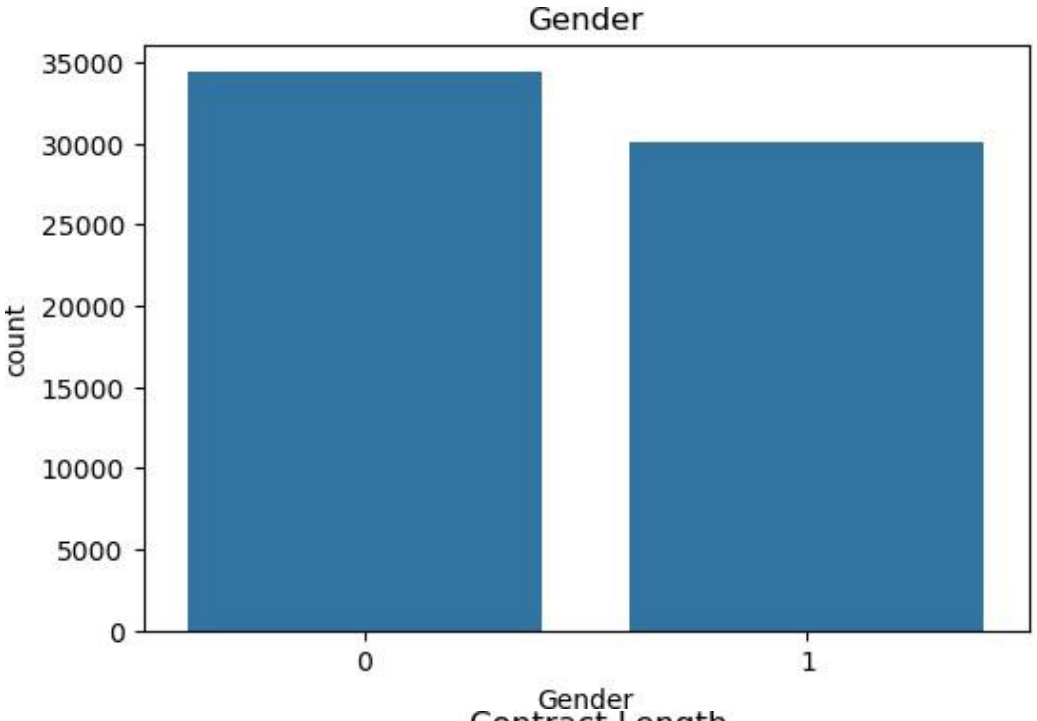
Total Spend 0

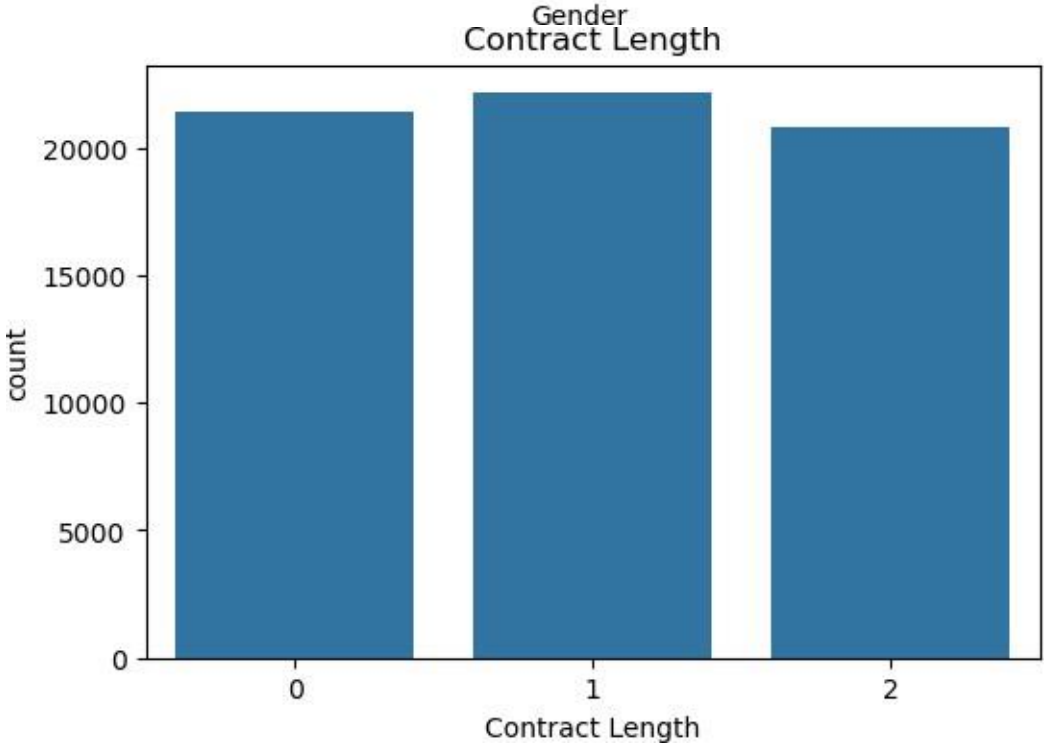
Last Interaction 0

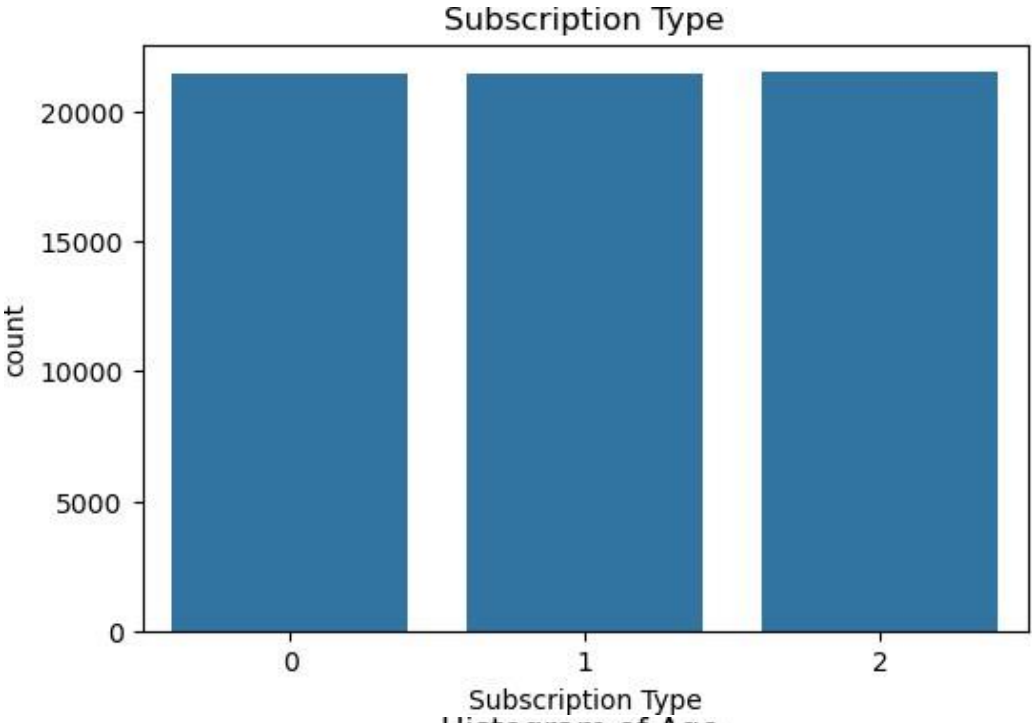
Churn 0 dtype: int64

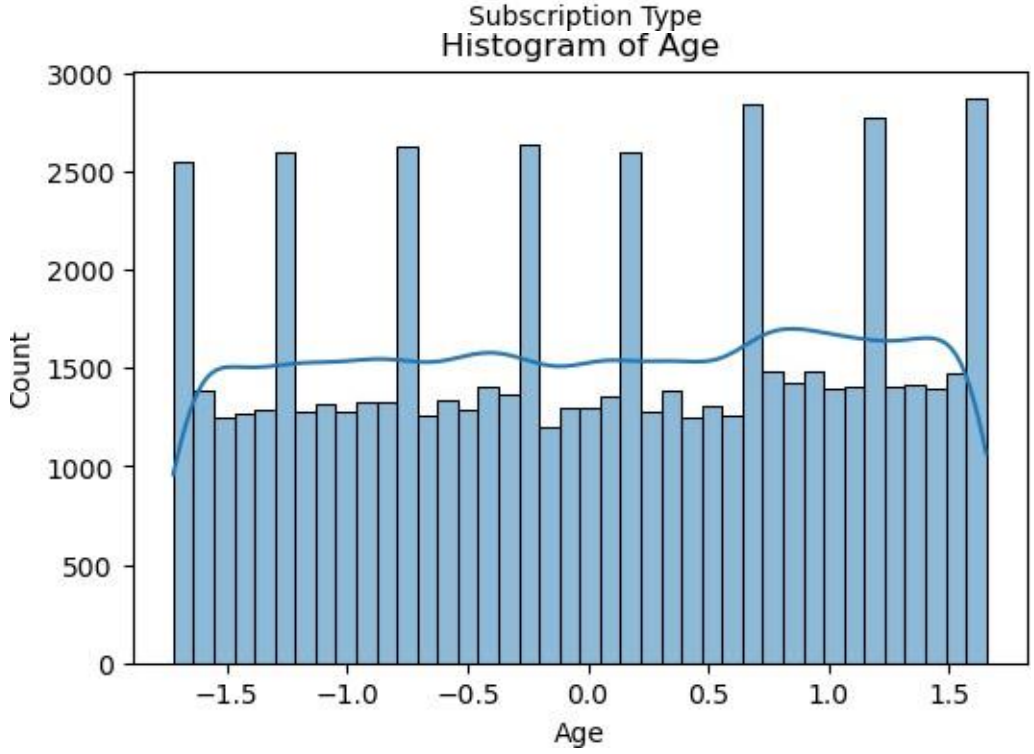

Correlation Heatmap
CustomerID 1 0.043 -0.086 0.1 -0.063 0.16 0.29 -0.0018 -0.011 -0.037 -0.0037 0.53
Age 0.043 1 0.0018 -0.0078 -0.038 0.005 -0.016 0.0045 9.3e-05 0.0065 -0.00015 0.063
Gender -0.086 0.0018 1 0.029 -0.0069 0.035 -0.059 0.0064 -0.0089 0.029 -0.00047 -0.16
Tenure 0.1 -0.0078 0.029 1 0.023 0.06 0.056 0.0042 0.00073 0.0095 0.0058 0.2
Usage Frequency -0.063 -0.038 -0.0069 0.023 1 -0.014 0.031 -0.00065 0.011 0.0015 -0.0092 -0.12
Support Calls 0.16 0.005 0.035 0.06 -0.014 1 0.064 -0.0032 -0.0031 0.022 0.0017 0.3
Payment Delay 0.29 -0.016 -0.059 0.056 0.031 0.064 1 -0.0015 0.0019 -0.031 -0.0081 0.56
Subscription Type -0.0018 0.0045 0.0064 0.0042 -0.00065 0.0032 -0.0015 1 -3.6e-05 0.0036 -0.0022 -0.0077
Contract Length -0.011 9.3e-05 -0.0089 0.00073 0.011 -0.0031 0.0019 -3.6e-05 1 0.0066 0.0039 -0.017
Total Spend -0.037 0.0065 0.029 0.0095 0.0015 0.022 -0.031 -0.0036 0.0066 1 -0.0077 -0.079
Last Interaction -0.0037 -0.00015 -0.00047 0.0058 -0.0092 0.0017 -0.0081 -0.0022 0.0039 -0.0077 1 -0.0028
Churn 0.53 0.063 -0.16 0.2 -0.12 0.3 0.56 -0.0077 -0.017 -0.079 -0.0028 1
CustomerID
Age
Gender
Tenure
Usage Frequency
Support Calls
Payment Delay
Subscription Type
Contract Length
Total Spend
Last Interaction
Churn
1.0
0.8
0.6
0.4
0.2
0.0

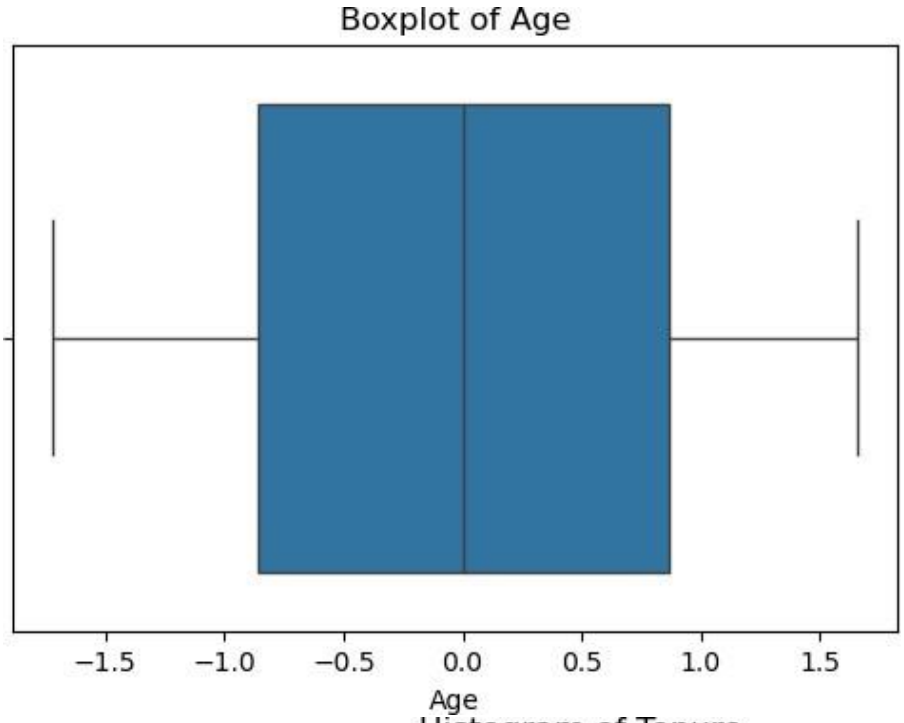

Churn
count
35000
30000
25000
20000
15000
10000
5000
0
0
1
Churn

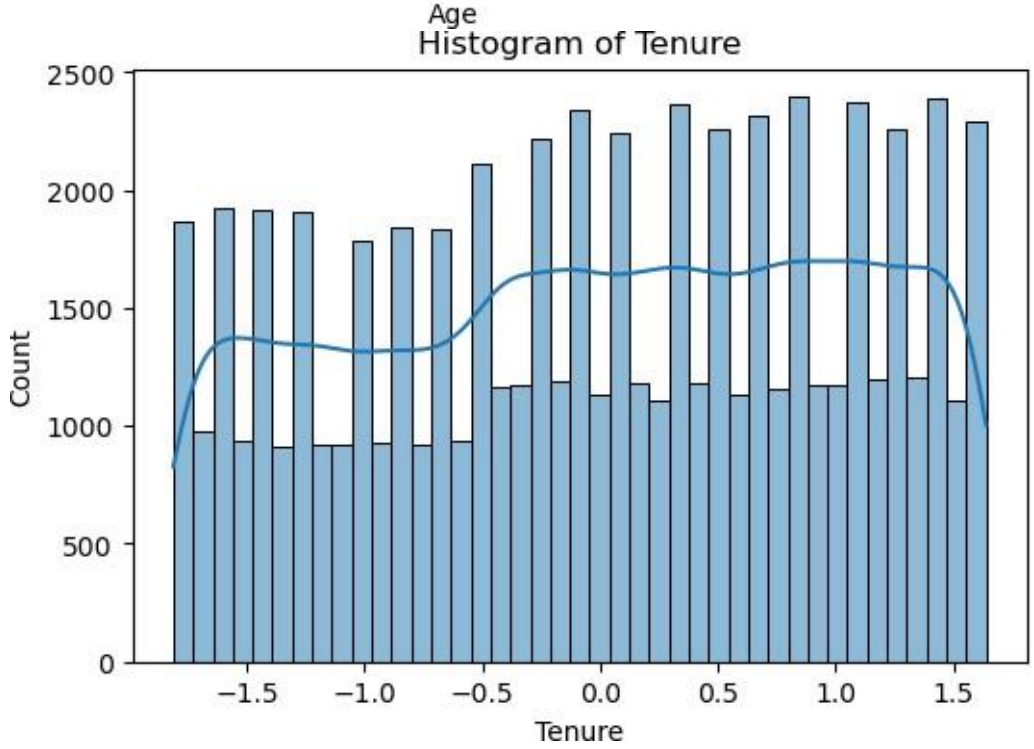

Gender
35000
30000
25000
20000
15000
10000
5000
0
count
0
1
Gender

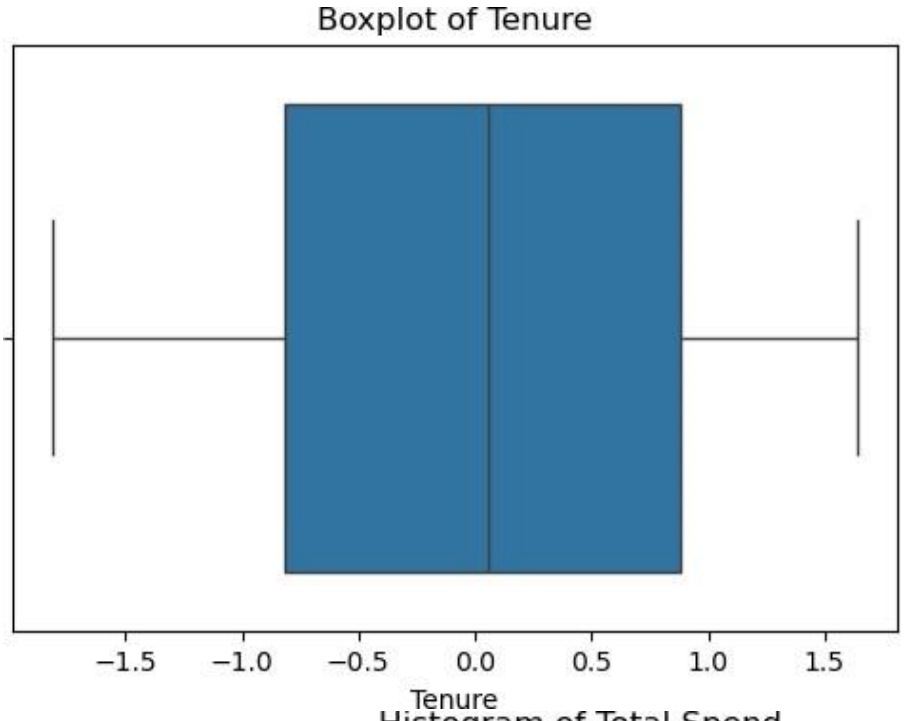

Contract Length
20000
15000
10000
5000
0
count
0
1
2
Contract Length

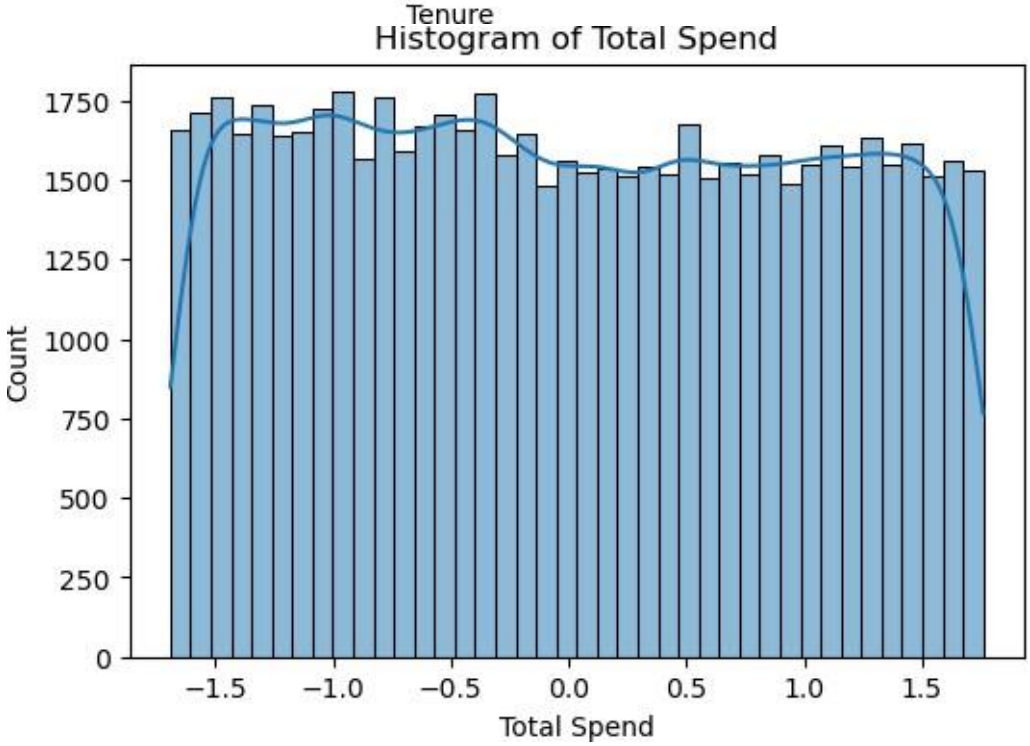

Subscription Type
20000
15000
10000
5000
0
count
0
1
2
Subscription Type

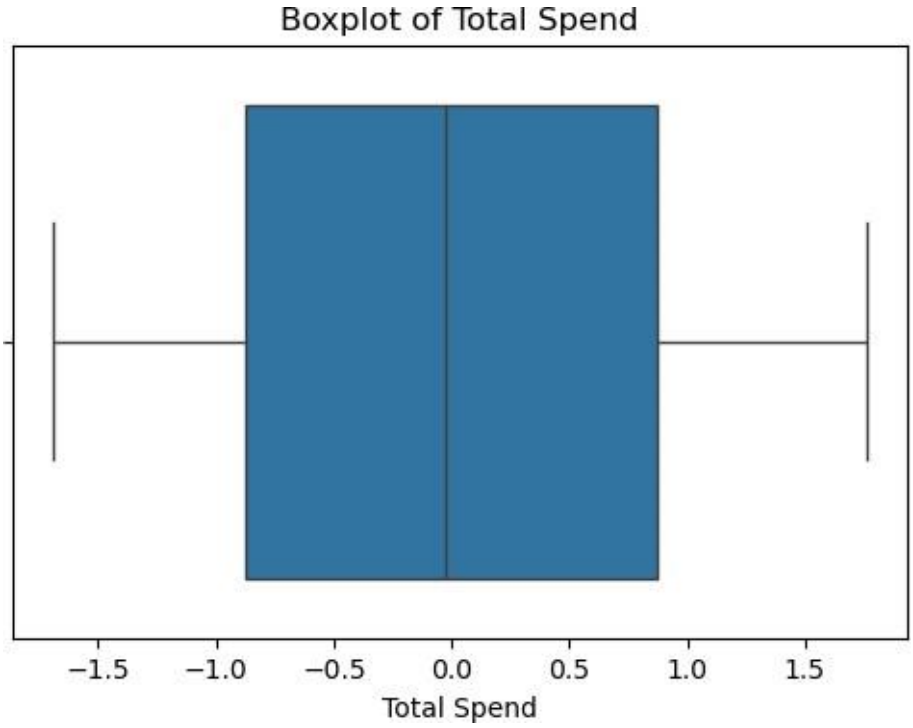

Histogram of Age
3000
2500
2000
1500
1000
500
0
Count
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
Age


Boxplot of Age
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
Age


Histogram of Tenure
2500
2000
1500
1000
500
0
Count
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
Tenure


Boxplot of Tenure
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
Tenure


Histogram of Total Spend
Count
0
250
500
750
1000
1250
1500
1750
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
Total Spend



Outliers using IQR

Age : 0

Tenure : 0

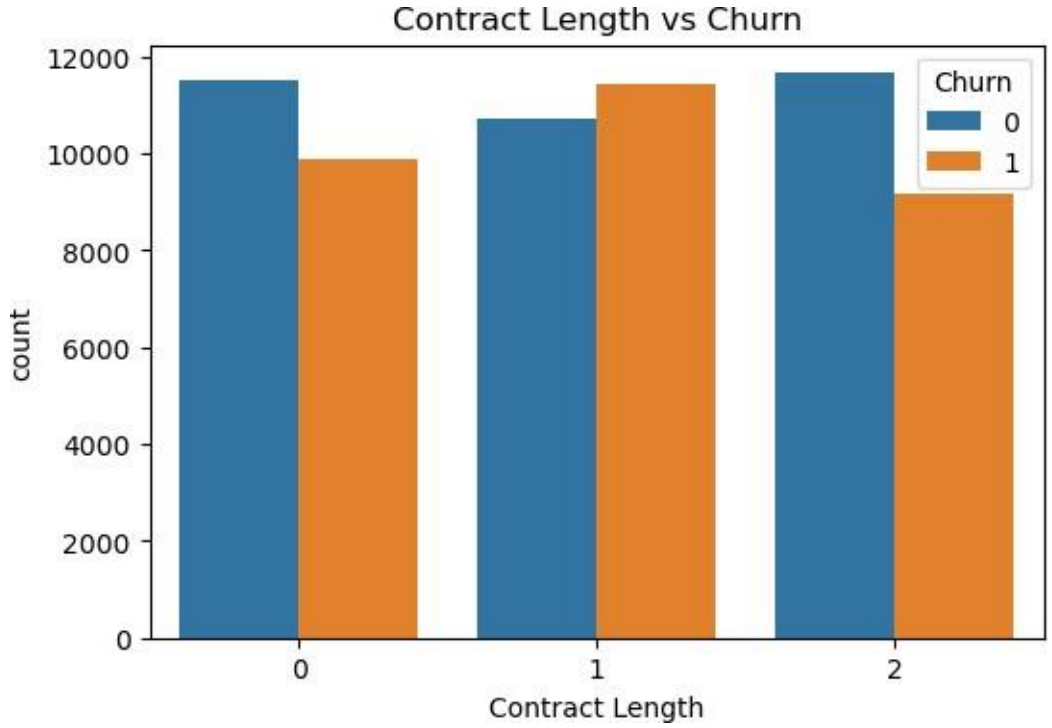
Usage Frequency : 0
Support Calls : 0

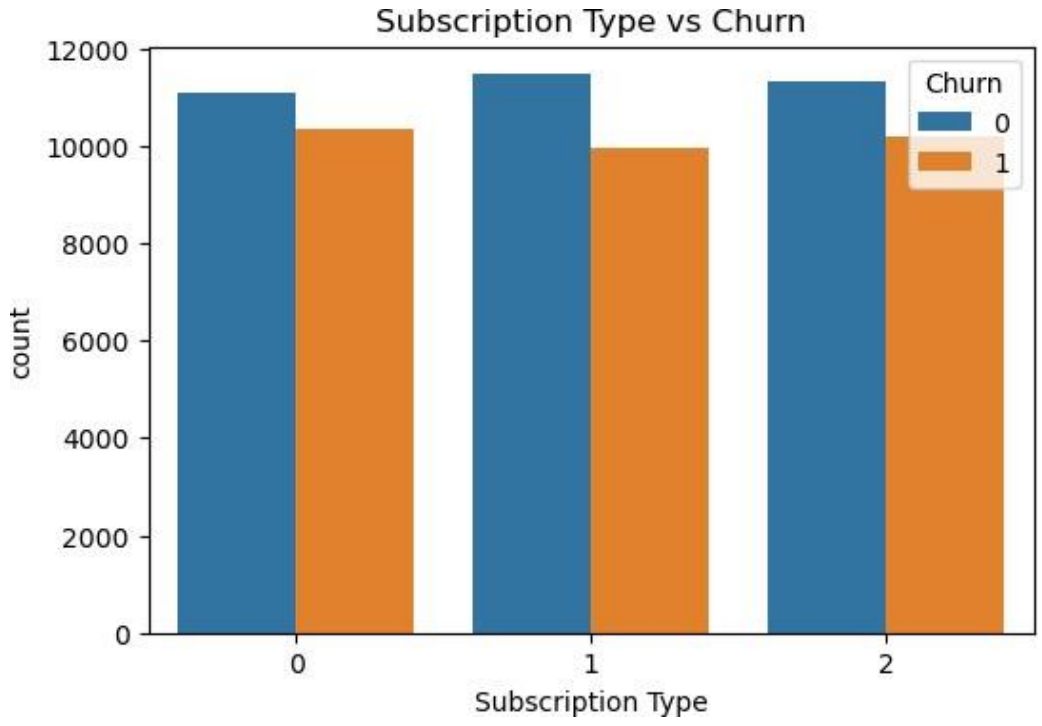
Payment Delay : 0

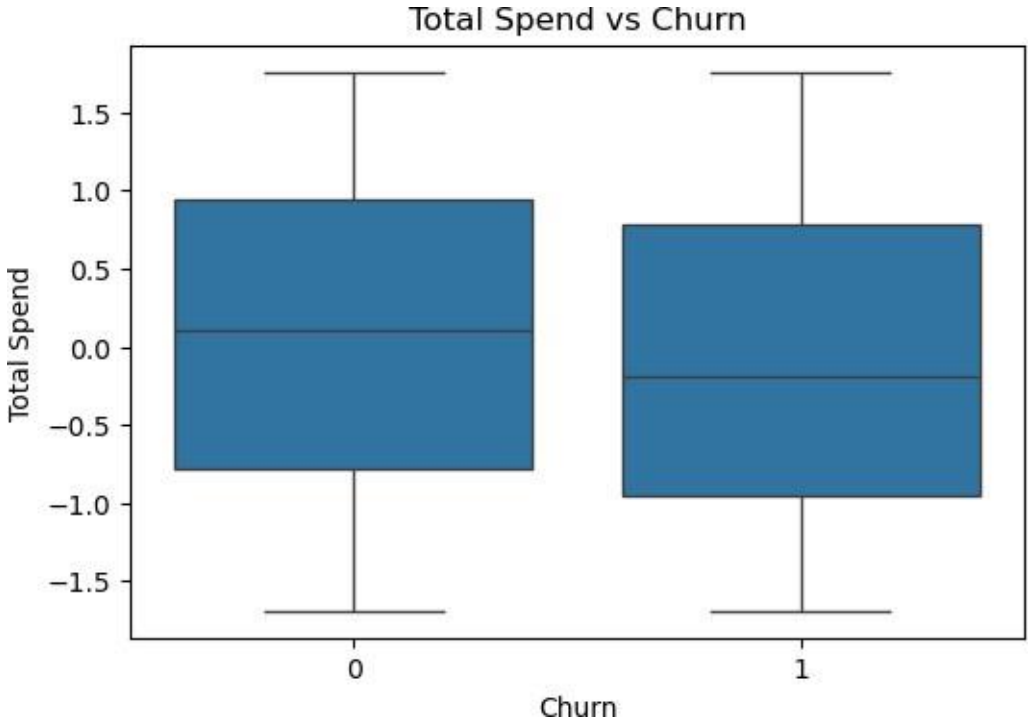
Total Spend : 0

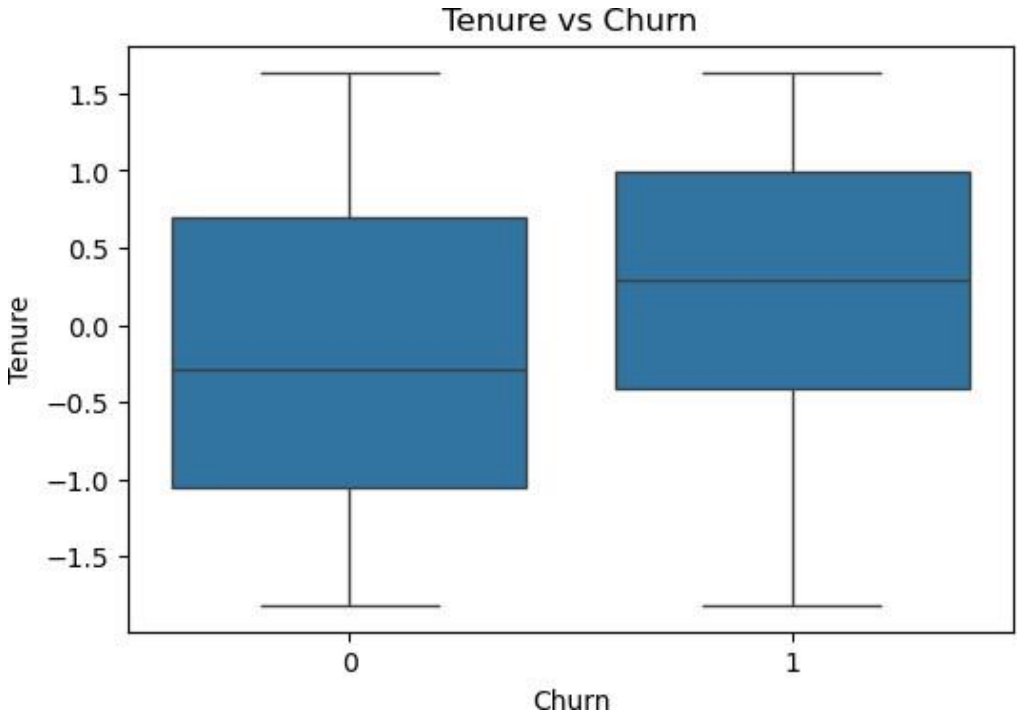
Last Interaction :

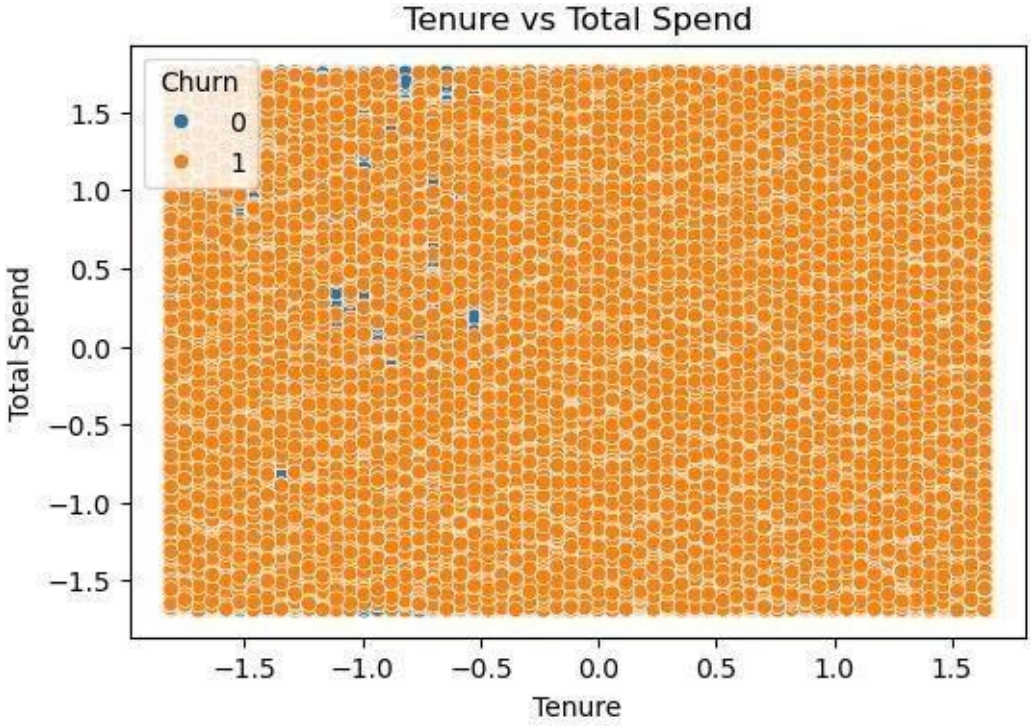
0


Contract Length vs Churn
12000
10000
8000
6000
4000
2000
0
count
0
1
2
Contract Length
Churn
0
1


Subscription Type vs Churn
12000
10000
8000
6000
4000
2000
0
count
0
1
2
Subscription Type
Churn
0
1


Total Spend vs Churn
Total Spend
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
0
1
Churn


Tenure vs Churn
Tenure
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
0
1
Churn



Contract Length with Highest Churn

Contract Length

0 0.462167

1 0.516087

2 0.440482

Name: Churn, dtype: float64

Subscription Type with Highest Churn

Subscription Type

0 0.482775

1 0.464964

2 0.473305

Name: Churn, dtype: float64

Average Tenure by Churn

Churn

0 -0.185304

1 0.205893

Name: Tenure, dtype: float64

Average Total Spend by Churn

Churn

0 0.074820

1 -0.083133

Name: Total Spend, dtype: float64

Correlation with Churn

Churn 1.000000

Payment Delay 0.557386

CustomerID 0.529832

Support Calls 0.304631

Tenure 0.195327

Age 0.063457

Last Interaction -0.002818

Subscription Type -0.007738

Contract Length -0.017098

Total Spend -0.078867

Usage Frequency -0.115098

Gender -0.164549

Name: Churn, dtype: float64

Training Data Shape : (51499, 11)

Testing Data Shape : (12875, 11)

Dataset Saved Successfully!
File Name : CustomerChurn_Preprocessed.csv

Learning Outcomes

After completing this experiment, students will be able to explore and preprocess datasets, handle missing values and duplicate records, encode categorical features, standardize numerical data, perform distribution, correlation, relationship, and outlier analysis, split the dataset into training and testing sets, and save the processed dataset for building machine learning models